2.6.2 Waveform Recognition Problem

Because of the elementary structure of the digit recognition problem, another, more complex, example was constructed. It is a three-class problem based on the waveforms $h_1(t)$, $h_2(t)$, $h_3(t)$ graphed in Figure 2.15.

Each class consists of a random convex combination of two of these waveforms sampled at the integers with noise added. More specifically, the measurement vectors are 21 dimensional: $x = (x_1, ..., x_{21})$. To generate a class 1 vector x, independently generate a uniform random number u and 21 random numbers $\varepsilon_1, ..., \varepsilon_{21}$ normally distributed with mean zero and variance 1. Then set

$$x_m = uh_1(m) + (1 - u)h_2(m) + \varepsilon_m, \; m = 1, ..., 21.$$

To generate a class 2 vector, repeat the preceding and set

$$x = uh_1(m) + (1 - u)h_3(m) + \varepsilon_m, \; m = 1, ...,21.$$

Class 3 vectors are generated by

$$x = uh_2(m) + (1 - u)h_3(m) + \varepsilon_m, \; m = 1, ..., 21.$$

Three hundred measurement vectors were generated using prior probabilities of $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, so there were approximately 100 per class. The data in the first five cases in each class are plotted in Figure 2.16. The tree was constructed using all questions of the form {Is $x_m \leq c$?} for c ranging over all real numbers and m = 1, ..., 21. The Gini splitting criterion (Section 2.5.2) was used, and the final tree selected by means of pruning and cross-validation.

This tree is shown in Figure 2.17. It has 11 terminal nodes. The resubstitution estimate of misclassification is .14; the cross-validation

estimate is .29; and the estimate of $R^*(T)$ gotten by using an independent test set of size 5000 is .28. Within each node in Figure 2.17 is the population, by class, of the node. Underneath each node is the question producing the split.

To illustrate how the splitting mechanism works, examine the first few splits. Figure 2.18 shows the waveforms $h_j(t)$ superimposed. The first split is $x_6 \leq 2.0$. The sixth coordinate is nearly at the peak of the first waveform. Since classes 1 and 2 both contain a random proportion of the first waveform, x_6 will be high for these two, except when the proportion is small or ε_6 is large and negative. This is reflected in the split with 109 out of the 115 class 3's going left (yes) and most of classes 1 and 2 going right (no).

Then the tree mechanism attempts to purify the left node by the split $x_{11} \leq 2.5$. At the eleventh coordinate the third waveform has its peak, so low values of x_{11} would tend to characterize class 1. The split sends 27 out of 36 class Is left, with most of classes 2 and 3 going right.

The right node resulting from the original split has almost equal proportions of classes 1 and 2 (64, 68). The mechanism attempts to separate these two classes by the split $x_{10} \leq 2.6$ near the peak of the third waveform. We would expect class 1 to be low at x_{10} and class 2 to be high. The resulting split confirms our expectations with most of class 1 going left and most of class 2 going right.

The numbers to the right of each terminal node are the normalized counts, by class, of the 5000 test sample vectors that were run down the tree and ended up in that terminal node. For comparability, the counts are normalized so that the total in the node equals the learning sample total.

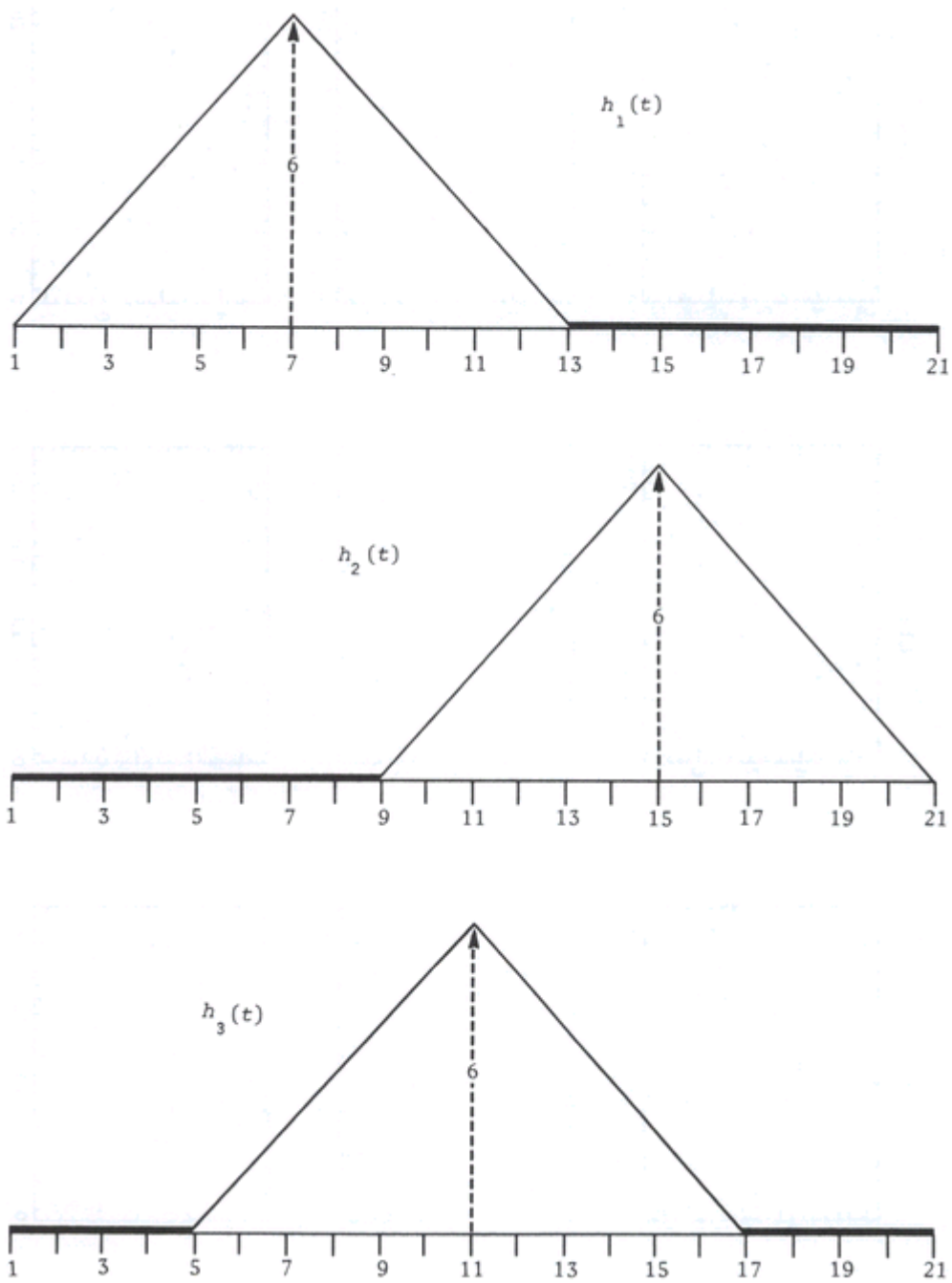
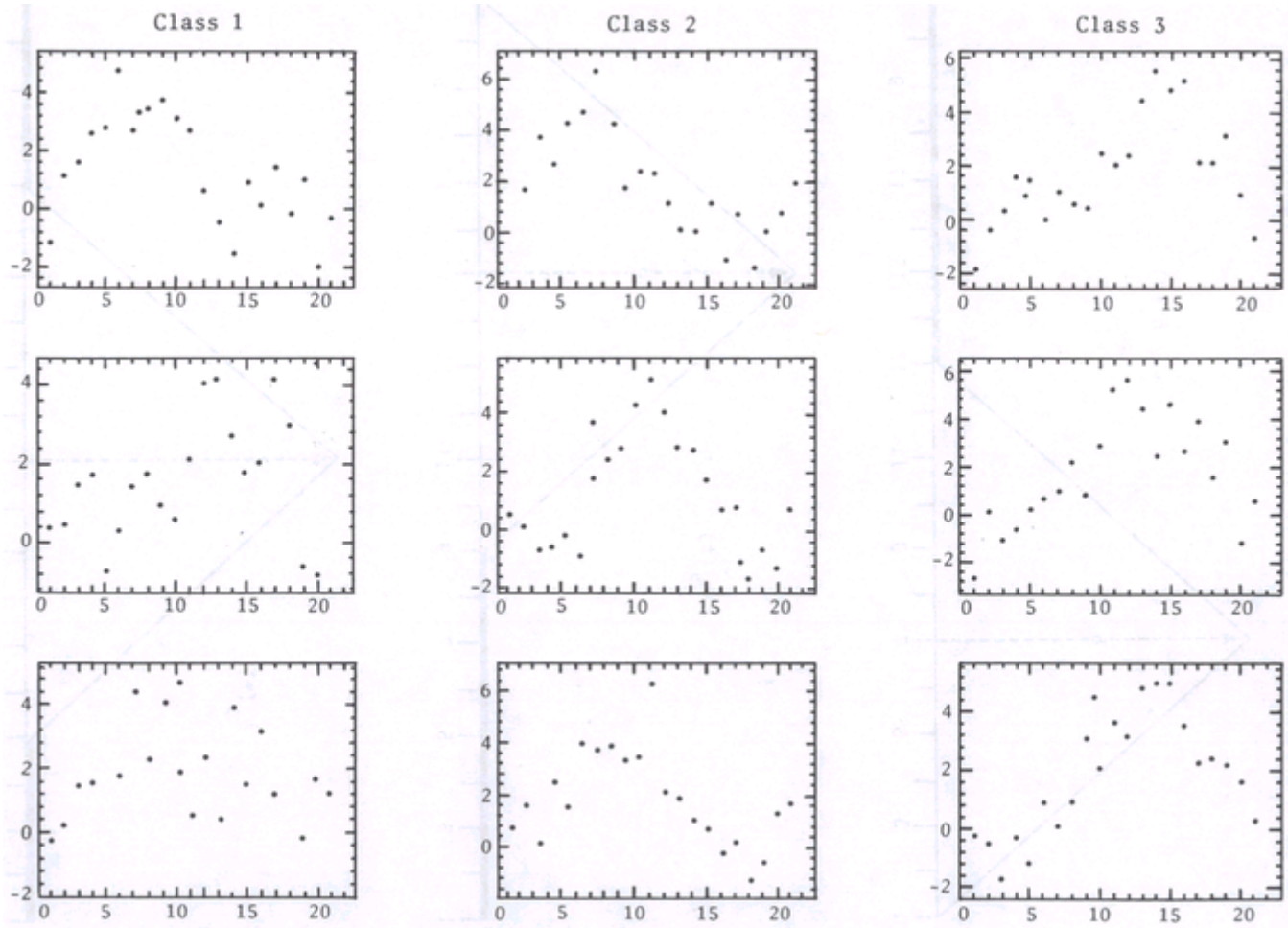


FIGURE 2.15 Waveforms.


Class 1
Class 2
Class 3

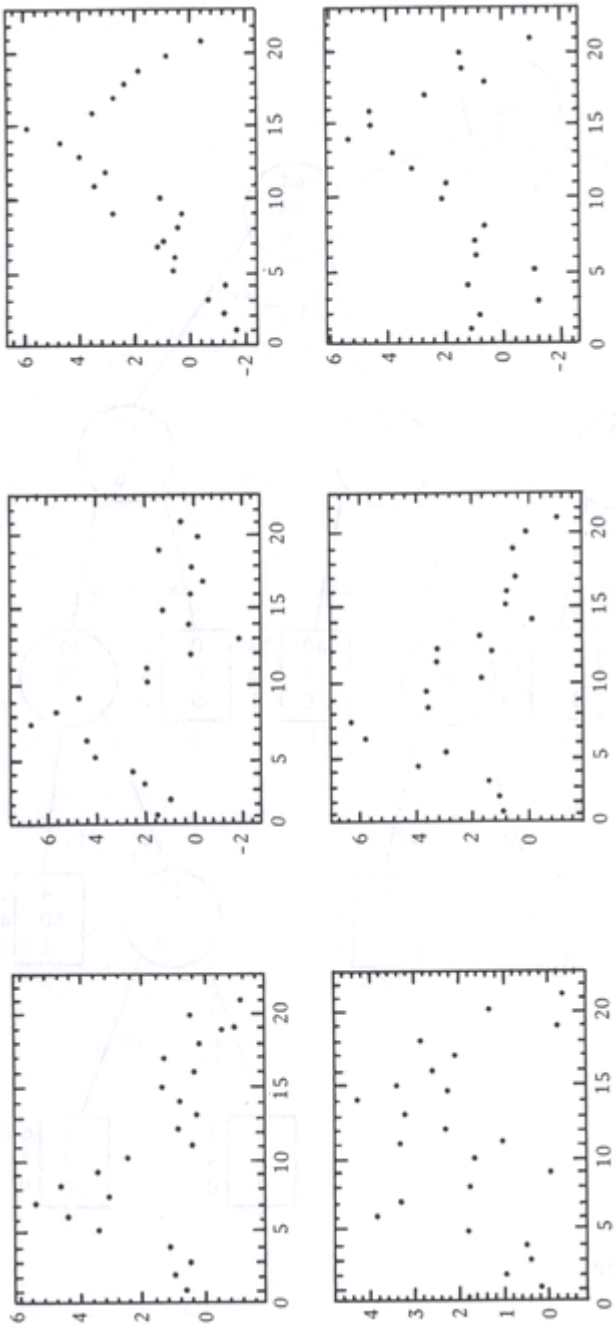


FIGURE 2.16

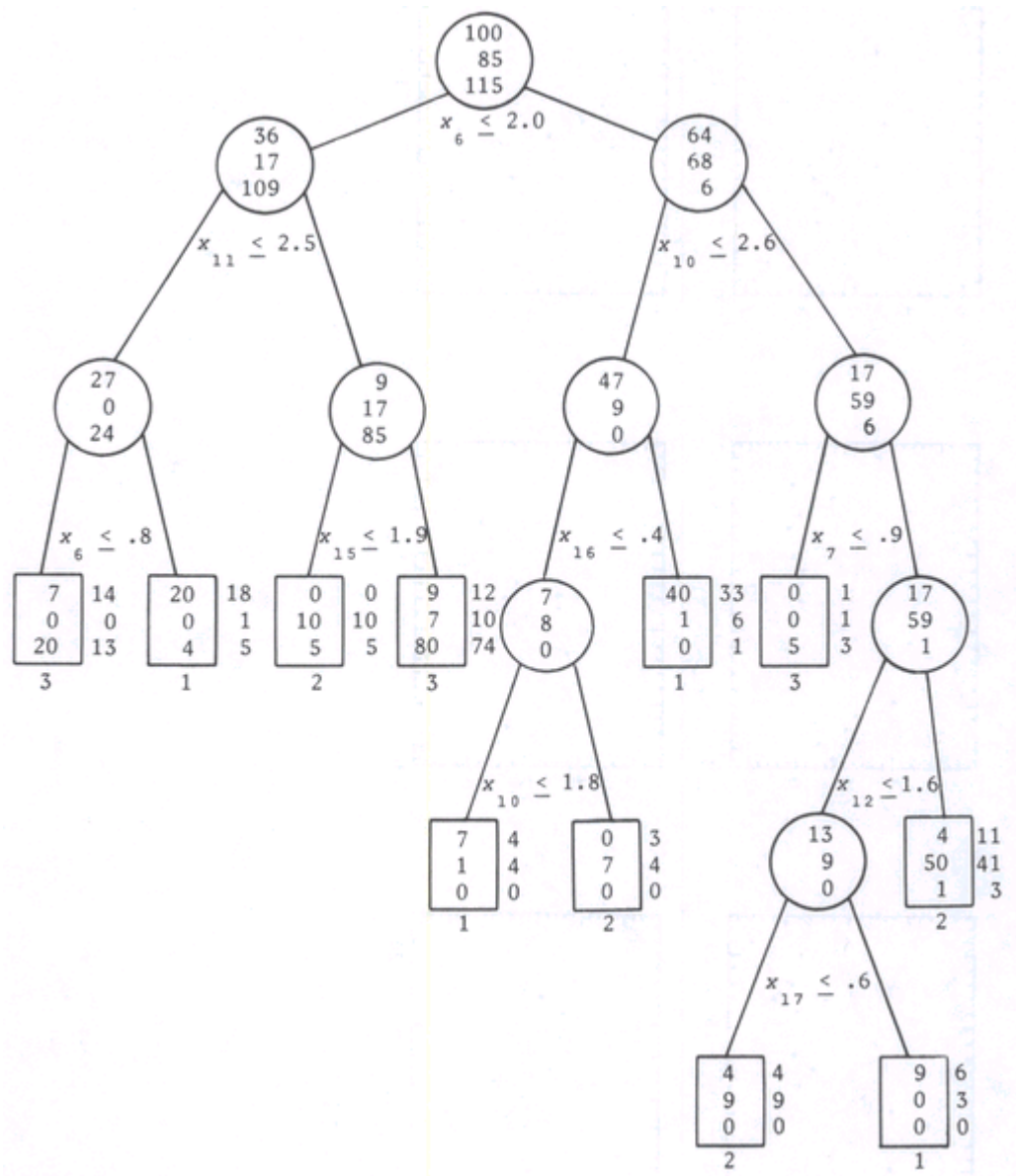


FIGURE 2.17 Waveform tree.

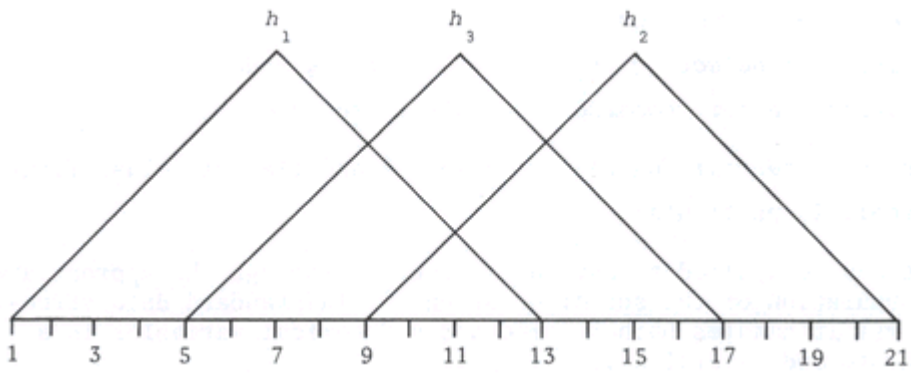


FIGURE 2.18

Comparing these two lists for the various terminal nodes, we can see that the learning sample node proportions can occasionally give quite misleading estimates. This is particularly true for the smaller nodes.

An analytic expression can be derived for the Bayes rule in this problem. Using this rule on a test sample of size 5000 gave an error rate of .14.

This example was selected to illustrate features of the tree construction procedure. That the tree classifier error rate is about twice the Bayes rate is an indication that single coordinate splits are not well suited to extracting the available information from this particular data set. The use of combinations of variables will be looked at in Section 5.2.